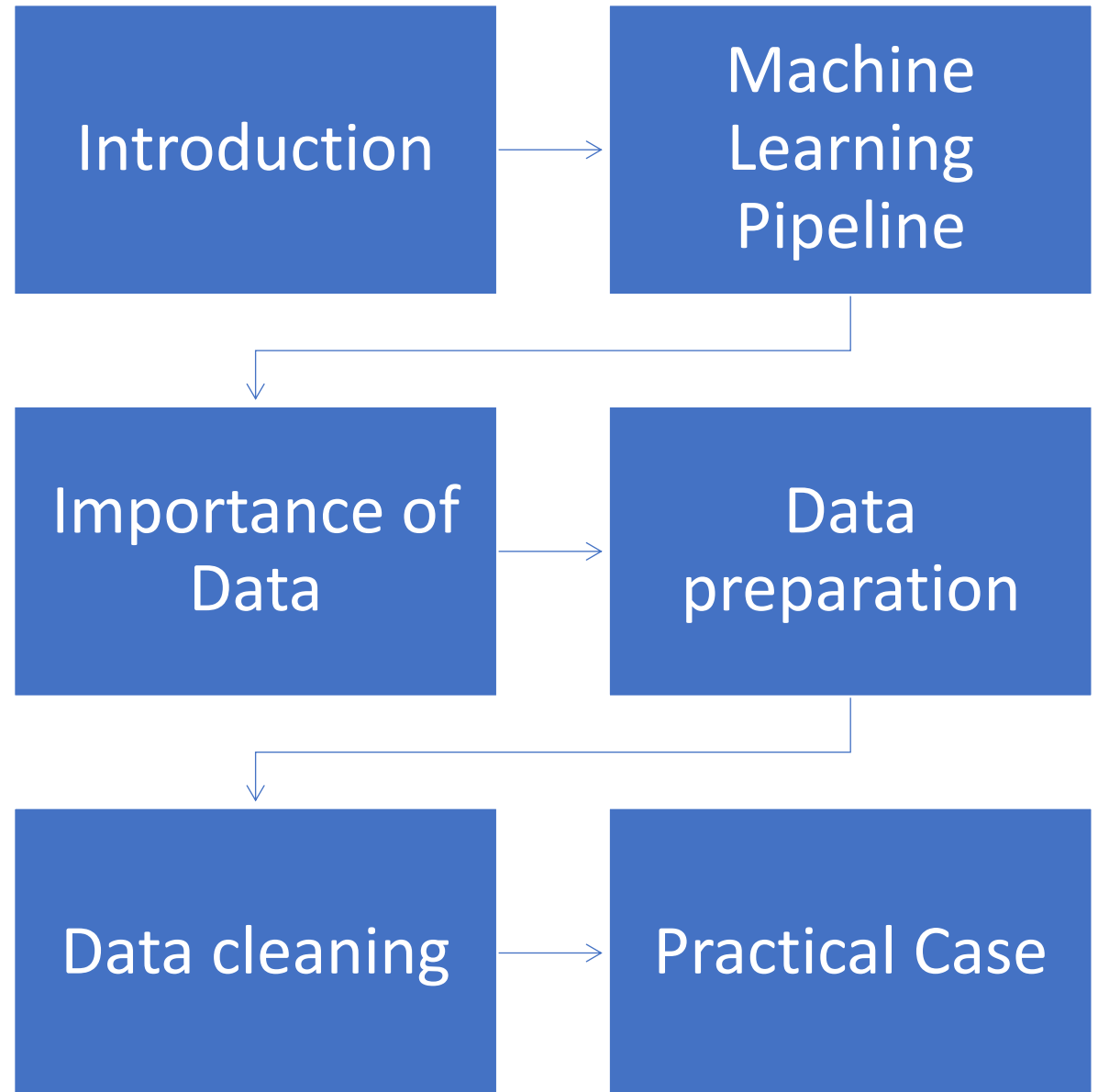


Interrupted times series analysis

Thursday, 08 September 2022

IBM Research Africa

Outlines



Introduction

- **What is a time series?**

When a variable is measured repeatedly over time, it is a time series.

- **What is Interrupted time series analysis (ITS)?**

ITS is a method of statistical analysis that follows a long-term period before and after an intervention point to assess the effects of the intervention.

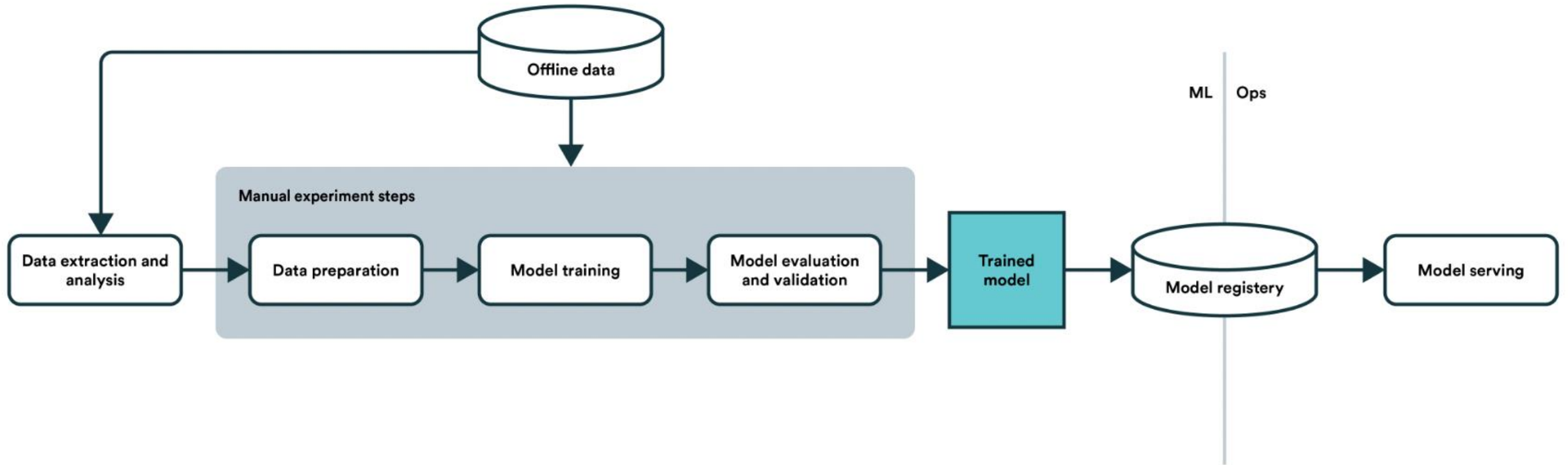
- **When to use?**

When you have a time series and want to understand how and if changes occurred in the series after an intervention, policy, or program was implemented for the entire population at one or more points in time.

Machine Learning Pipeline

A machine learning pipeline is a way to codify and automate the workflow required to generate a machine learning model. Machine learning pipelines are made of several sequential steps that range from data extraction and preprocessing to model training and deployment.

Machine learning pipeline workflow



Importance of Data

- Data is useful information that you collect to support organizational decision-making and strategy.
- By allowing you to measure and take action, an effective data system can enable your organization to improve the quality of people's lives.
- An important distinction to make is the difference between data and information.

Type of Data

- **Qualitative data:** Qualitative data is descriptive in nature and expressed in terms of emotions rather than numbers. Because qualitative data analysis represents data, it cannot be counted or quantified. They are phrases or labels used to define specific characteristics or attributes. Examples of qualitative data are gender, name, country, etc.
- **Quantitative data:** The value of data expressed in numbers or figures is known as quantitative data. Any quantifiable information that can be used for mathematical calculations and statistical analysis for the purpose of making real-world decisions based on the results is called quantitative data. Examples of quantitative data are cost, weight, height, age, number, etc.

Data preparation

Data preparation is the process of cleaning and transforming raw data before processing and analysis. It is an essential step prior to processing, often involving restructuring data, correcting data that has been incorrectly recorded, and merging datasets to enrich them.

The benefits of data preparation :

- Fix errors quickly
- Produce high quality data
- Make better business decisions

Data preparation steps

Gather

- Finding the right data

Discover

- Learn about the data
- Understand what happened before

Cleanse

- Removing outliers
- Filling in missing values

Transform and Enrich

- Updating the format or value entries
- adding and connecting data with other related information

Store

- Store the data into a third party application

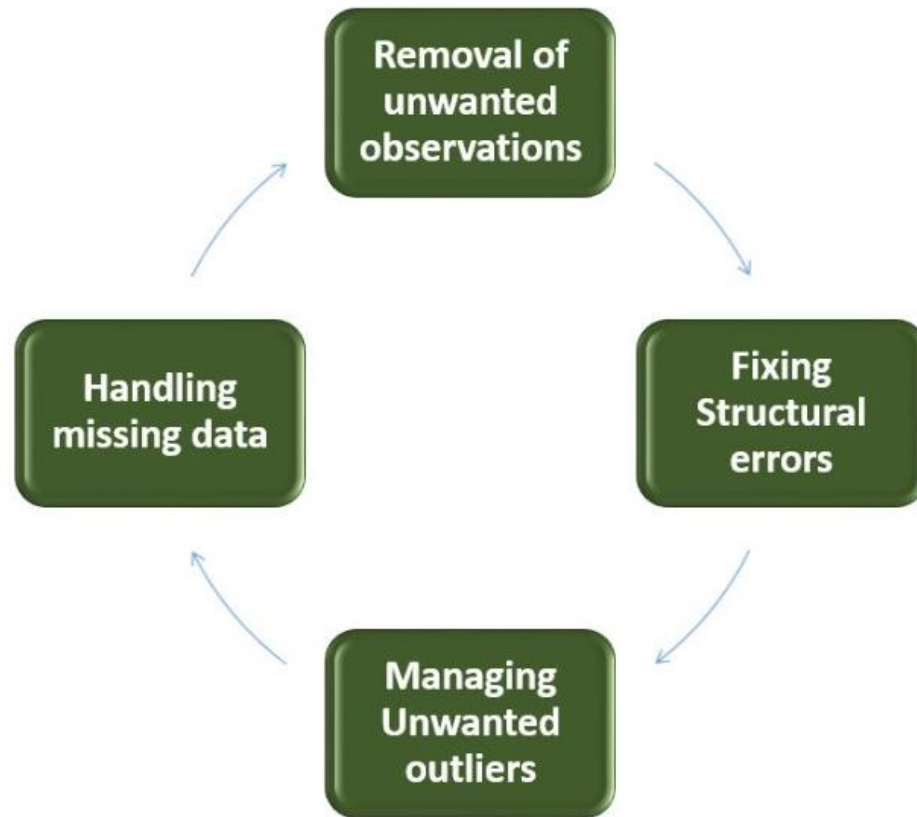
Why Data cleaning?

Data cleaning is the process of correcting or removing incorrect, corrupted, improperly formatted, duplicated, or incomplete data in a data set.

This is certainly not the most sophisticated part of machine learning and, at the same time, there are no hidden tricks or secrets to discover. However, the success or failure of a project relies on good data cleaning. If the data is incorrect, the results and algorithms are not reliable, even if they appear correct.



Steps involved in Data Cleaning



Steps involved in Data Cleaning: Removal of unwanted observations

This includes removing duplicate, redundant, or irrelevant features from your data set. Duplicate observations often occur during data collection and irrelevant features are those that do not match the specific problem you are trying to solve.

S.N°	First Name	Last Name	Title	Company
1	Mary	Sue	Senior Marketing Manager	ABC Ltd.
2	Janet	Martin	Marketing Executive	ABC Ltd.
3	Bryan	Oscar	SEO Manager	ABC Ltd.
4	Jude	Taylor	Marketing Manager	ABC Ltd.
5	Mary S	Sue	Senior Marketing Manager	ABC Ltd.

Steps involved in Data Cleaning: Fix structural errors

During measurement, data transfer or similar situations, errors can occur, these are called structural errors.

- Spelling mistakes or extra punctuation
- Inconsistencies in formatting
- The same attribute with a different name
- Data entries

Steps involved in Data Cleaning:

Managing outliers

- Outliers are entries whose numerical amounts differ significantly from the other similar entries. The most common causes of outliers in a data set are data entry errors, measurement errors and sampling errors.

How to detect outliers?

- Sorting method
- Data visualization method
- Statistical tests (z scores)

Steps involved in Data Cleaning: Managing outliers

- **Sorting Method:** This involves looking for extremely low or extremely high values by sorting your quantitative variables.

Example:

Let say you have a data set of 8 values:

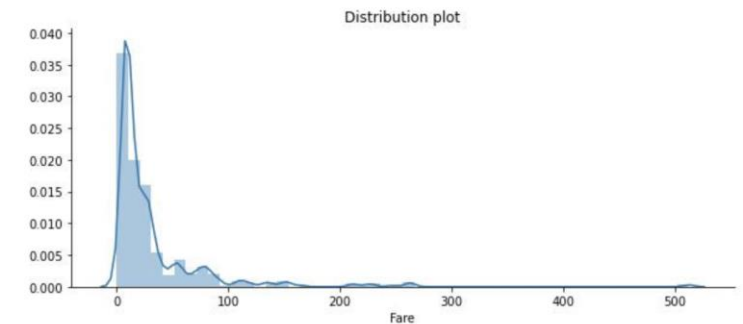
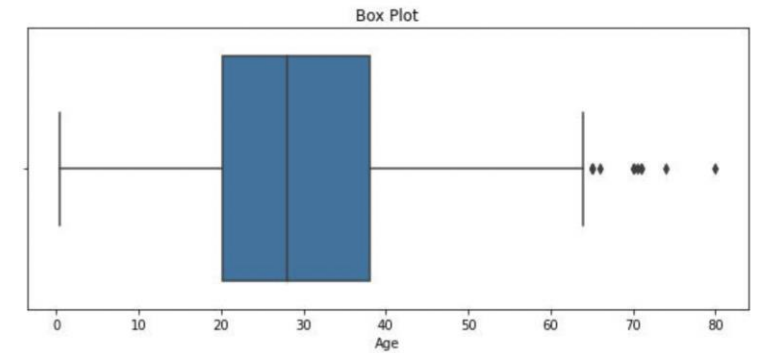
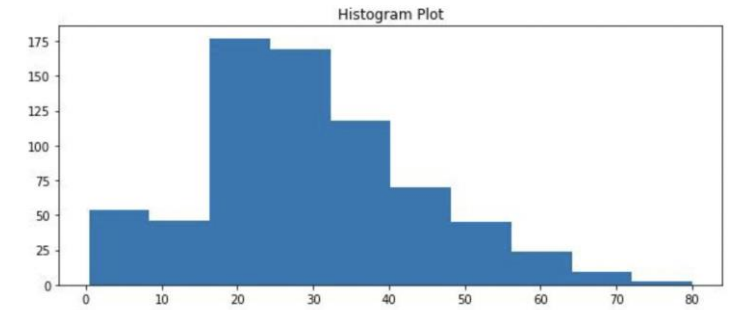
18	14	20	105	10	12	23	9
----	----	----	-----	----	----	----	---

You sort your values from low to high and identify extreme values

9	10	12	14	18	20	23	505
---	----	----	----	----	----	----	-----

Steps involved in Data Cleaning : Managing outliers

- **Data visualization method:** You can use software to visualize your data using a box plot, histogram, distribution plot, scatter plot, to see the distribution of the data at a glance.



Steps involved in Data Cleaning: Managing outliers

- **Statistical tests (z scores)**

Z-score formula	Explanation
$z = \frac{x - \mu}{\sigma}$	• x = individual value • μ = mean • σ = standard deviation

If a value has a sufficiently high or low z-score, it can be considered an outlier. As a general rule, values with a z-score greater than 3 or less than -3 are often considered outliers.

Steps involved in Data Cleaning: Managing outliers

So far, we have learned how to detect outliers. The main question is what do we do with outliers?

Here are some of the methods for dealing with outliers

- Trimming/removing the outlier
- Quantile-based floor and ceiling
- Imputing the mean/median

Steps involved in Data Cleaning: Handle missing data

Missing data are, unfortunately, unavoidable in poorly designed data collection procedures. They must be identified and addressed as quickly as possible.

Missing values

PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	male	22	1	0	A/5 21171	7.25		S
2	1	1	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	female	26	0	0	STON/O2. 3101282	7.925		S
4	1	1	female	35	1	0	113803	53.1	C123	S
5	0	3	male	35	0	0	373450	8.05		S
6	0	3	male		0	0	330877	8.4583		Q

Steps involved in Data Cleaning: Handle missing data

To handle missing values, you can drop the record if a variable used in the model is missing, drop the variable or impute the data.

- 1) **Deleting rows:** This method is commonly used to deal with null values. Here, we delete a particular row if it has a null value for a particular feature and a particular column if it has more than 70-75% missing values.
- 2) **Replace by mean/median/mode:** This strategy can be applied to a feature that contains numerical data, such as the age of a person or the price of a ticket. We can calculate the mean, median or mode of the feature and replace it with the missing values.
- 3) **Assign a unique category:** This is for categorical characteristics, which have a defined number of possibilities, like gender, for example. Since they have a defined number of classes, we can assign another class for the missing values.

ANY
QUESTIONS?

[dreamstime.com](#)